Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

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Introduction

Theory framework

Prediction forward model

The prediction forward model, or forward model, is the underlying mapping between a model's parameters, controls and initial states to its observable features.

Denoting by $\mathbf{m}$ the vector of model parameters, and by $\mathbf{u}$ the set of controls applied on the field, and by $\mathbf{d}$ the reservoir production within the time frame of the controls, one can write

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}; \mathbf{x}_0) \tag{1}$$

with $\mathbf{x}_0$ the initial value of reservoir states (usually omitted).

Data features

In this work, the features inside $\mathbf{d}$ are the following well measurements:

▷ oil and water production rates	(OPR and WPR)	[m^3/day]
▷ water injection rate	(WIR)	[m^3/day]
▷ bottomhole pressure	(BHP)	[Bar]
▷ water cut percentage	(WCTP)	[%]

Field oil production total

Field oil production total (FOPT, SI unit of m^3) is also relevant to this work, despite not being part of $\mathbf{d}$.

The key assumption behind the forward model is that it is deterministic. Implementation wise, g consists of evaluating a set of equations at each gridblock of the model.

Each evaluation evolves the model a little further in time, until the scheduled time period is appreciated, concluding the process.

Black-oil equations are the most common implementations of the forward model [2] due to their fair simplicity and accuracy. The fluid is assumed composed of three liquid phases, water, oil and gas.

A given phase α's component mass A_α satisfies the conservation equation

$$\frac{\partial}{\partial t}\left(\phi_{\text{ref}} A_\alpha\right) + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0, \tag{2}$$

where

$$\begin{aligned} A_w &= m_\phi b_w s_w, & \mathbf{u}_w &= b_w \mathbf{v}_w, & \text{(3a)} \\ A_o &= m_\phi (b_o s_o + r_{og} b_g s_g), & \mathbf{u}_o &= b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g, & \text{(3b)} \\ A_g &= m_\phi (b_g s_g + r_{go} b_o s_o), & \mathbf{u}_g &= b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o. & \text{(3c)} \end{aligned}$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K}(\nabla p_\alpha - \rho_\alpha \mathbf{g}). \tag{4}$$

Finally, the following closure equations are imposed:

$$s_w + s_o + s_g = 1, \qquad p_{c,ow} = p_o - p_w, \qquad p_{c,og} = p_o - p_g. \tag{5}$$

Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, free of observation error. Data is separated into a history period and a validation period, so that

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_h \\ \mathbf{d}_v \end{bmatrix} \tag{6}$$

with $\mathbf{d}_h$ the ground-truth history data, and $\mathbf{d}_v$ the ground-truth validation data.

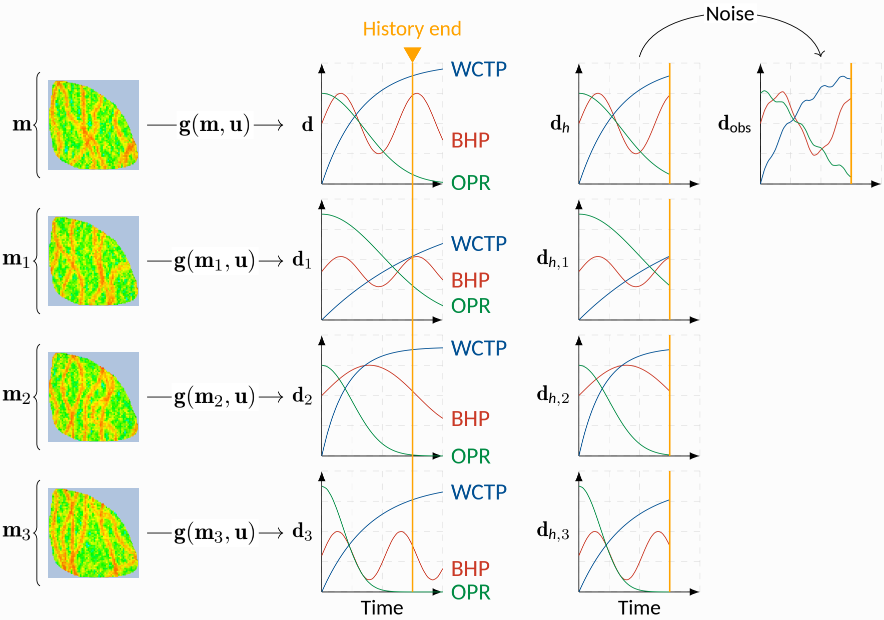
Ground-truth data come from the forward model,

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}). \tag{7}$$

and observed data, from a corruption of $\mathbf{d}_h$:

$$\mathbf{d}_{\text{obs}} = \mathbf{d}_h + \mathcal{N}(\mathbf{0}, \mathbf{C}_e), \tag{8}$$

where $\mathbf{C}_e$ is the noise covariance matrix.


History end
Noise
$\mathbf{m}$
$\mathbf{g}(\mathbf{m}, \mathbf{u})$
$\mathbf{d}$
WCTP
BHP
OPR
$\mathbf{d}_h$
$\mathbf{d}_{\mathrm{obs}}$
$\mathbf{m}_1$
$\mathbf{g}(\mathbf{m}_1, \mathbf{u})$
$\mathbf{d}_1$
$\mathbf{d}_{h,1}$
$\mathbf{m}_2$
$\mathbf{g}(\mathbf{m}_2, \mathbf{u})$
$\mathbf{d}_2$
$\mathbf{d}_{h,2}$
$\mathbf{m}_3$
$\mathbf{g}(\mathbf{m}_3, \mathbf{u})$
$\mathbf{d}_3$
$\mathbf{d}_{h,3}$
Time
Time

DSI-ESMDA can be conceptulized as a Bayesian approach to the forecasting problem, aiming to estimate the underlying conditional probability distribution of predictions given the reference data.

The DSI-ESMDA problem

Find the posterior distribution

$$p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}_{\text{obs}} \mid \mathbf{d}^*)p(\mathbf{d}^*). \tag{9}$$

where $\mathbf{d}^*$ maximizes $p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}})$.

The algorithm returns samples from this posterior distribution, outputing a range of values rather than a single value. While this makes evaluating the algorithm less straightforward, it allows for quantification of uncertainty.

Ensemble Smoothers share this approach at a conceptual level, but they perform a single maximization. This leads to poor matches to real data if assumptions on the likelihood $\mathcal{L}$ are not met.

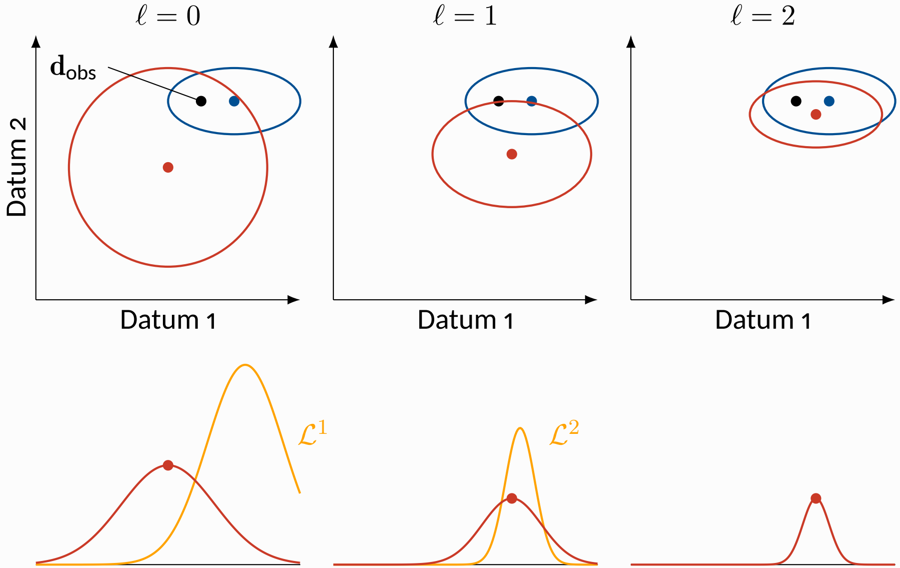
Multiple Data Assimilation (MDA) mitigates this problem by making N_a optimizations with resampled reference data:

$$p^1(\mathbf{d}^* \mid \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^1_{\text{obs}} \mid \mathbf{d}^*)p(\mathbf{d}^*) \tag{10}$$

$$p^2(\mathbf{d}^* \mid \mathbf{d}^2_{\text{obs}}, \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^2_{\text{obs}} \mid \mathbf{d}^*)p^1(\mathbf{d}^* \mid \mathbf{d}^1_{\text{obs}}) \tag{11}$$

$$\vdots \tag{12}$$

$$p^{N_a}(\mathbf{d}^* \mid \mathbf{d}^{N_a}_{\text{obs}}, \ldots, \mathbf{d}^1_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}^{N_a}_{\text{obs}} \mid \mathbf{d}^*)p^{N_a-1}(\mathbf{d}^* \mid \mathbf{d}^{N_a-1}_{\text{obs}}, \ldots, \mathbf{d}^1_{\text{obs}}) \tag{13}$$


$\ell = 0$
$\ell = 1$
$\ell = 2$
$\mathbf{d}_{\rm obs}$
Datum 2
Datum 1
Datum 1
Datum 1
$\mathcal{L}^1$
$\mathcal{L}^2$

MDA makes the following suppositions on the likelihoods:

- $\mathcal{L}^\ell$ is Gaussian
- $\mathcal{L}^\ell$ has covariance $\mathbf{C}_e^\ell = \alpha^\ell \mathbf{C}_e$, $\alpha^\ell > 0$

This leads to

$$\mathcal{L}^\ell(\mathbf{d}_{\text{obs}}^\ell \mid \mathbf{d}^*) \propto \exp\left(-\frac{1}{2}(\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*)^\top \frac{\mathbf{C}_e^{-1}}{\alpha^\ell}(\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*)\right), \tag{14}$$

$\mathbf{H}$ a matrix that extracts $\mathbf{d}_h^*$ from $\mathbf{d}^*$.

Equivalence to ES

For Gaussian likelihoods, MDA is equivalent to ES if

$$\sum_{\ell=1}^{N_a} \frac{1}{\alpha^\ell} = 1, \tag{15}$$

highlighting the importance of having inflation factors on the covariance.

The priors are also supposed Gaussian:

$$p^{\ell}(\mathbf{d}^* \mid \mathbf{d}^{\ell}_{\text{prior}}) \propto \exp\left(-\frac{1}{2}(\mathbf{d}^* - \mathbf{d}_{\text{prior}})^{\top}(\mathbf{C}^{\ell}_{p})^{-1}(\mathbf{d}^* - \mathbf{d}_{\text{prior}})\right), \tag{16}$$

where

- $\mathbf{d}^{\ell}_{\text{prior}}$ is the prediction with highest belief of occurring prior to assimilating $\mathbf{d}^{\ell}_{\text{obs}}$
- $\mathbf{C}^{\ell}_{p}$ is the prior covariance matrix (i.e., $\mathbf{d}^* \sim \mathcal{N}(\mathbf{d}^{\ell}_{\text{prior}}, \mathbf{C}^{\ell}_{p})$)

The maximum *a posteriori* (MAP) estimate of the ℓ-th assimilation becomes

$$\mathbf{d}_{\text{map}}^{\ell+1} = \mathbf{d}_{\text{map}}^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}}^{\ell} - \mathbf{d}_{h,\text{map}}^{\ell}), \tag{17a}$$

$$\mathbf{K}^{\ell} = \mathbf{C}_p^{\ell}\mathbf{H}^{\top}(\mathbf{H}\mathbf{C}_p^{\ell}\mathbf{H}^{\top} + \alpha^{\ell}\mathbf{C}_e)^{-1} \tag{17b}$$

with $\mathbf{d}_{\text{map}}^{0} = \mathbf{d}_{\text{prior}}^{0}$.

Using an ensemble of N_e model realizations, one can estimate the prior covariances with $\Delta\mathbf{D} = \begin{bmatrix}\mathbf{d}_1 - \bar{\mathbf{d}} & \cdots & \mathbf{d}_{N_e} - \bar{\mathbf{d}}\end{bmatrix}$:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}}\mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \tag{18a}$$

$$\mathbf{K}^{\ell} = \Delta\mathbf{D}^{\ell}\Delta\mathbf{D}_h^{\ell}(\Delta\mathbf{D}_h^{\ell}\Delta\mathbf{D}_h^{\ell} + \alpha^{\ell}\mathbf{C}_e)^{-1}. \tag{18b}$$

$\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}}\mathbf{e}_j^{\ell}$, with $\mathbf{e}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_e)$, is a way of sampling $\mathbf{d}_{\text{obs}}^{\ell} \sim \mathcal{N}(\mathbf{d}_{\text{obs}}, \alpha^{\ell}\mathbf{C}_e)$.

Covariance localization, or simply localization, is an *ad hoc* procedure employed to mitigate the effects of small ensembles [1, section 7.5].

The influence a datum has on another datum is weighted in a inverse manner to the distance between the points.

Localization is applied to the Kalman-like gain, so that the base equation becomes:

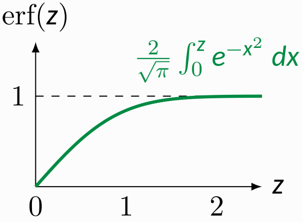
$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}}\mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \tag{19}$$

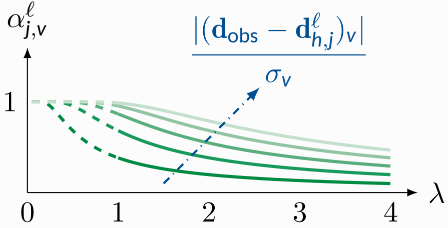
This work introduces gating α on the innovation vector:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})[\boldsymbol{\alpha}_j^{\ell} \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^{\ell})], \tag{20a}$$

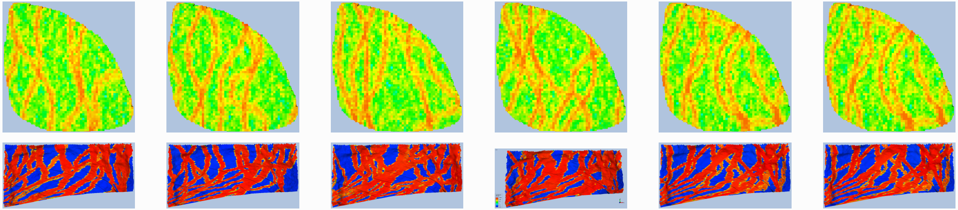
$$\alpha_{j,v}^{\ell} = \operatorname{erf}\left(\frac{1}{\lambda\sqrt{2}} \frac{|(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^{\ell})_v|}{\sigma_v}\right). \tag{20b}$$

The gating recovers variance loss in the posterior ensemble by diminishing Kalman-like updates on reasonably matched data. The smaller the mismatch of a measurement relative to the noise at that datum, the smaller the update.



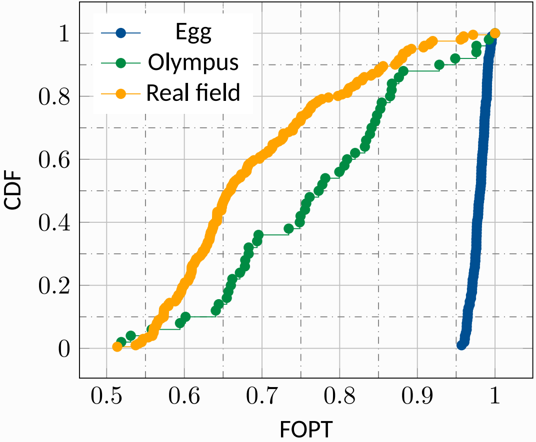


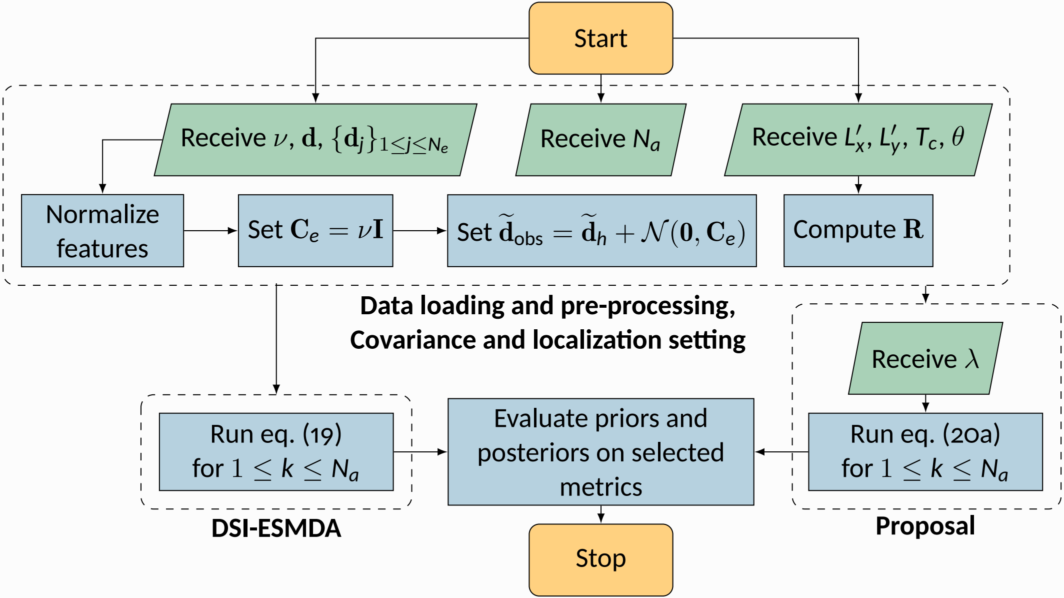
Methods



Property	Egg	Olympus	Real field	Unit
Rock compressibility	0.0	1.42×10^{-10}	-	1/Pa
Water compressibility	1.0×10^{-10}	3.97×10^{-10}	-	1/Pa
Oil dynamic viscosity	5.0×10^{-3}	$2.8\text{-}3.5 \times 10^{-3}$	-	Pa·s
Water dynamic viscosity	1.0×10^{-3}	0.398×10^{-3}	-	Pa·s
Well diameter	20	19	-	cm
Wells	8(P) 4(I)	11(P) 7(I)	16(P) 8(I)	-
Active cells	18,533	192,750	-	-
Cell size	$8\times8\times4$	$50\times50\times3$	-	m×m×m
Ensemble size	101	50	201	-

The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.




Start
Receive ν, $\mathbf{d}$, $\{\mathbf{d}_j\}_{1\leq j\leq N_e}$
Receive N_a
Receive L'_x, L'_y, T_c, θ
Normalize features
Set $\mathbf{C}_e = \nu\mathbf{I}$
Set $\widetilde{\mathbf{d}}_{obs} = \widetilde{\mathbf{d}}_h + \mathcal{N}(\mathbf{0}, \mathbf{C}_e)$
Compute $\mathbf{R}$
Data loading and pre-processing, Covariance and localization setting
Receive λ
Run eq. (19) for $1 \leq k \leq N_a$
Evaluate priors and posteriors on selected metrics
Run eq. (20a) for $1 \leq k \leq N_a$
DSI-ESMDA
Proposal
Stop

Four metrics are considered on the evaluation of posterior/prior distributions [1, subsection 9.4.2]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch.

Denoting an ensemble by $\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 & \cdots & \mathbf{y}_{N_e} \end{bmatrix}$, these metrics are defined as:

$$\mathrm{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1}\left(\min(\mathbf{Y})_i \leq d_{h,i} \leq \max(\mathbf{Y})_i\right) \tag{21}$$

$$\mathrm{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1}\left(\min(\mathbf{Y})_i \leq d_{v,i} \leq \max(\mathbf{Y})_i\right) \tag{22}$$

$$\mathrm{MSE}(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \overline{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \overline{\mathbf{y}}), \quad \mathbf{C} = \Delta\mathbf{Y}\Delta\mathbf{Y}^\top + \mathbf{C}_e \tag{23}$$

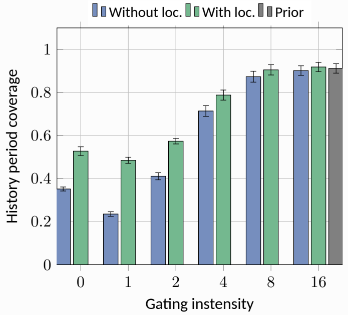
$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}) \tag{24}$$

Reservoir	N_e	Number of time steps	Time step size	Simulation life		History split
Egg	100	120	30 days	3600 days	(9.8 years)	60%
Olympus	49	20	1 calendar year	7305 days	(20 years)	60%
Real field	200	510	1 calendar month	12785 days	(35 years)	60%*

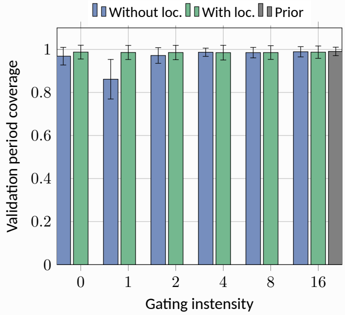
Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

Reservoir	Well features used		Number of wells		Total features
	Producer	Injector	Producer	Injector	
Egg	OPR, WPR	BHP	4	8	16
Olympus	OPR, WPR	WIR	11	7	29
Real field	BHP, WCTP	BHP	16	8	39 (a WCTP was constant)

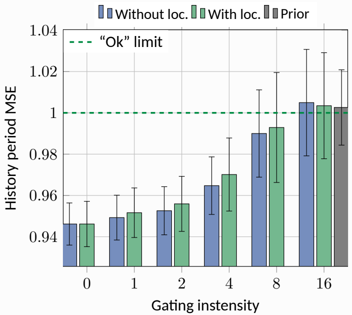
Results



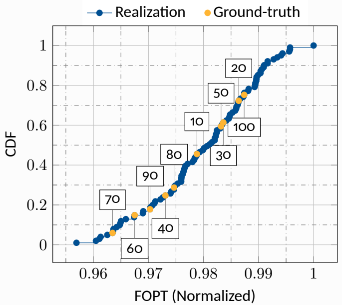
Without loc.
With loc.
Prior
History period coverage
0
0.2
0.4
0.6
0.8
1
0
1
2
4
8
16
Gating instensity



Without loc.
With loc.
Prior
Validation period coverage
0
0.2
0.4
0.6
0.8
1
0
1
2
4
8
16
Gating instensity

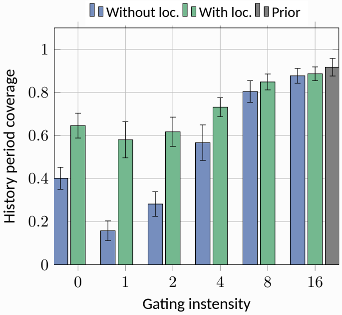


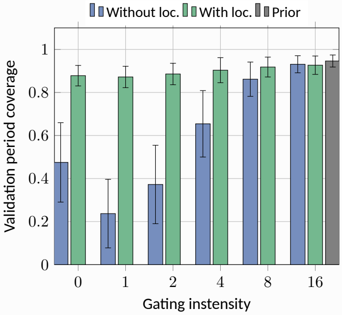
Without loc.
With loc.
Prior
"Ok" limit
History period MSE
0.94
0.96
0.98
1
1.02
1.04
0
1
2
4
8
16
Gating instensity

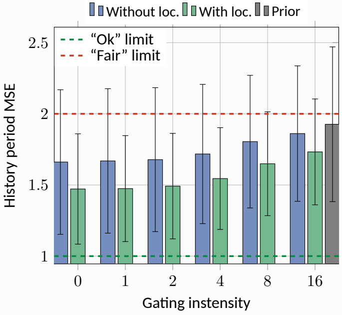


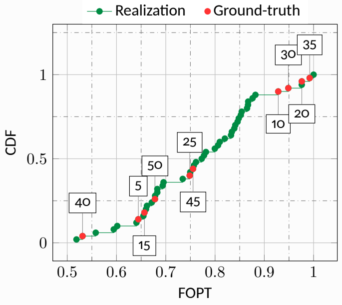
Realization
Ground-truth
CDF
0
0.2
0.4
0.6
0.8
1
20
50
10
100
80
30
90
70
40
60
0.96
0.97
0.98
0.99
1
FOPT (Normalized)

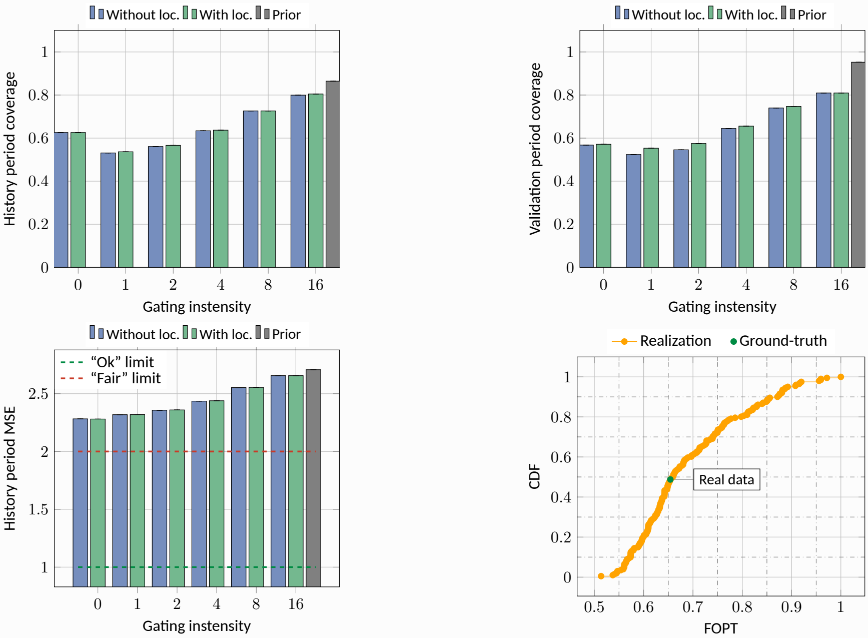
Results ► Olympus case










Without loc.
With loc.
Prior
History period coverage
Gating instensity
0
1
2
4
8
16
Validation period coverage
Gating instensity
History period MSE
"Ok" limit
"Fair" limit
Gating instensity
Realization
Ground-truth
Real data
CDF
FOPT

OC, MSE and mismatch increase

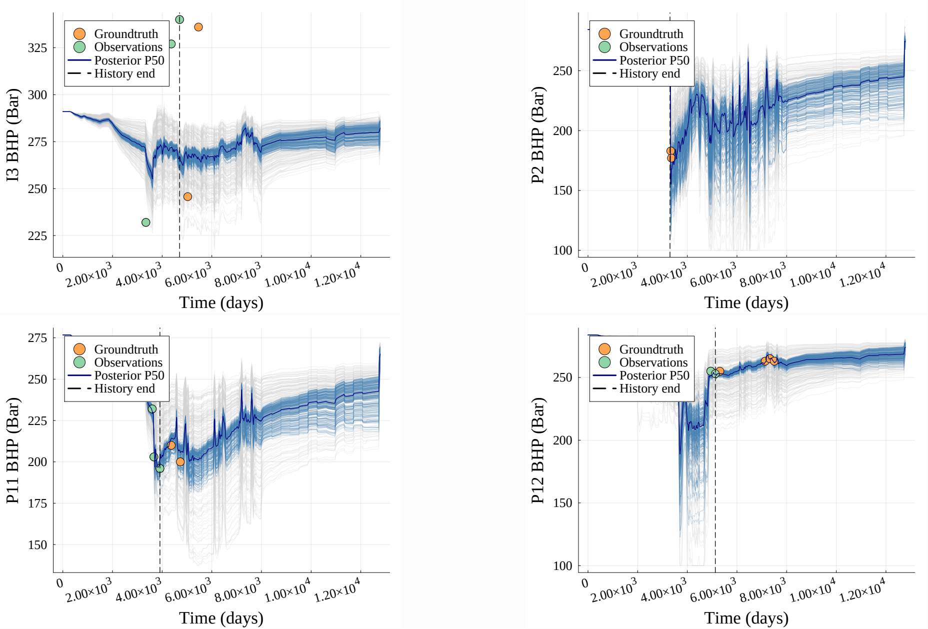
Increase in gating intensity tended to increase observation coverage, mean squared error, and data mismatch, albeit with case-dependent sensitivities.

- Gating reduces posterior variance loss, so increase in coverages, but also MSE and mismatch, are expected as gating intensifies.
- Egg standard deviations are lesser than Olympus on two accounts: reservoir complexity, and ground-truth representativeness. Olympus' ground-truths were somewhat biased.

Mixed effects of localization

Localization yielded mixed results: for the Egg and real field cases, effects on coverage and MSE were minimal, especially for intense enough gatings, while the effect in the Olympus case was drastic.

- **Egg case**: narrow FOPT range and roughly uniform sampling of ground-truths leave no ground-truth unrepresented, reduced dimentionality help with covariance estimation.
- **Olympus case**: much more complex while also having a much smaller ensemble and a lot more features, aggravating covariance estimation. Localization was conceived exactly for such situations, so it is not suprising it was effective here.
- **Real field case**: peak complexity and dimentionality, but also ensemble size. Localization may have inadequately adjusted, and some features are not well-represented.


Groundtruth
Observations
Posterior P50
History end
I3 BHP (Bar)
P2 BHP (Bar)
P11 BHP (Bar)
P12 BHP (Bar)
Time (days)
0
2.00×10^3
4.00×10^3
6.00×10^3
8.00×10^3
1.00×10^4
1.20×10^4

Mild and small gating intensities versus standard DSI-ESMDA

Mild gating intensities (2 in the Egg and Olympus cases, 4 in the real field case) produced similar results to the standard DSI-ESMDA for same localization configuration, while small gating (1 for the Egg case, 1 and 2 for the Olympus and real field cases) had worse coverages than the standard DSI-ESMDA.

- Gating intensity “reverts” the posterior to the prior. If the posterior given by the standard DSI-ESMDA is somewhere in that continuum, then it stands to reason that Gated DSI-ESMDA could approximate it simply by tuning the gating intensity.
- Since Gated DSI-ESMDA does away with observation data resampling, small gatings possibly underestimate uncertainty (i.e., likelihood covariance) at least when compared to standard DSI-ESMDA, producing posteriors that are not well-matched to data.

Conclusion

Bibliography

[1] Alexandre Anozé Emerick. *Ensemble Data Assimilation Applied to Geological Reservoir Models*. 1st ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, June 2025. ISBN: 978-65-88763-29-2.

[2] Atgeirr Flø Rasmussen et al. "The Open Porous Media Flow reservoir simulator". In: *Computers & Mathematics with Applications* 81 (2021). Development and Application of Open-source Software for Problems with Numerical PDEs, pp. 159–185. ISSN: 0898-1221. DOI: `https://doi.org/10.1016/j.camwa.2020.05.014`.

Gated DSI-ESMDA

Reducing variance loss in posteriors

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"It's hard to make predictions, specially about the future"

— Yogi Berra